

DATA PRODUCT CARD

PJATK • Big Data Course • Two-Page Template

1. PRODUCT OVERVIEW

Product name: UK Property Sales Insights
Owner / Team: Dominika Lutek
Business domain: Real Estate / Finance
Primary users: Real Estate Analysts, Investors, Data Scientists
Version / update date: v1.0 / 2026-05-07

AT A GLANCE

Format: DuckDB Table / dbt Models
Refresh: Continuous (Streaming)
Access: Public GitHub Repository
Quality metrics: Completeness (City), Uniqueness (ID), Validity (Price)
Status: Production-ready

2. PURPOSE & VALUE

This data product addresses the challenge of analyzing massive property transaction datasets (over 31 million records) in real-time. It supports investment decision-making by providing cleaned, deduplicated data on prices and locations, enabling the identification of market trends and liquidity in specific cities. The system seamlessly integrates historical records with new transactions arriving through a modern streaming pipeline.

3. DATA SOURCES

List the main source systems or files used to build this product.

Source	Type	Refresh
UK Land Registry (PP-Complete)	File (CSV)	Daily
Redpanda Stream	Streaming	Continuous

4. HOW TO ACCESS THIS PRODUCT

- Where is the product publishedPublic GitHub Repository
- Direct link: <https://github.com/readmebooks/projekt-nieruchomosci.git>
- How to download or query itClone the repository, activate the virtual environment (venv), and run the Dagster pipeline or query the nieruchomosci_uk.db file directly using a DuckDB client.
- Example file/table name: stg_uk_sales (Silver layer) or avg_price_by_city (Gold layer)

5. QUALITY SNAPSHOT

Metric	Value
Completeness (City)	100.00%
Uniqueness (Transaction ID)	100.00%
Validity (Price > 0)	100.00%
Row count	31,004,536 records

6. HIGH-LEVEL FLOW

Summarize the pipeline in one line:

Source	Bronze	Silver	Gold	Product
UK Land Registry CSV → bronze_stream_sales → stg_uk_sales (Silver) → avg_price_by_city (Gold) → Data Marketplace				

Detailed Contract & Operating Information

7. SCHEMA OVERVIEW

List the main fields that a consumer needs to understand.

Field name	Type	Description	Example
id	VARCHAR	Unique transaction identifier	{2A289E9F-6BB5-CDC8}
price	INTEGER	Final sale price in GBP	250000
sale_date	DATE	Date of the transaction	1995-03-24
city	VARCHAR	City or town name	LONDON
county	VARCHAR	Administrative county	GREATER LONDON
sale_year	BIGINT	Year extracted from the sale date	1995

8. REFRESH & OPERATIONS

Refresh frequency: Continuous (every 30 seconds)
Update trigger: Dagster Sensor monitoring Redpanda
Storage location: Local DuckDB file / GitHub Repo
Expected batch size / volume: ~50 rows per streaming batch
Dependencies:
Source Data: UK Land Registry "PP-Complete" CSV file (Full historical load).
Streaming Source: Redpanda / Kafka topic named uk_property_sales.
Infrastructure: Docker Desktop (required to run Redpanda/Kafka containers).
Environment: Python 3.12+ with Dagster and dbt-duckdb libraries installed.
Storage: Local file system access for the nieruchomosci_uk.db file.

9. EXAMPLE USAGE

Business question:
What is the average property price in each city?
Example query / code / steps:

```
SELECT city, AVG(price)
FROM stg_uk_sales
GROUP BY 1
ORDER BY 2 DESC
LIMIT 10;
```

10. LIMITATIONS & RISKS

- Geographical Scope:** Data is limited to England and Wales (excludes Scotland and NI).
- Hardware Requirements:** Processing the full 31M records efficiently requires at least 16GB of RAM.
- Latency:** The sensor check interval is approximately 30 seconds.

11. GOVERNANCE & CONTACT

Owner: Dominika Lutek
Contact: Teams
Access policy: Public for class
Marketplace location: Data Marketplace folder

Feedback